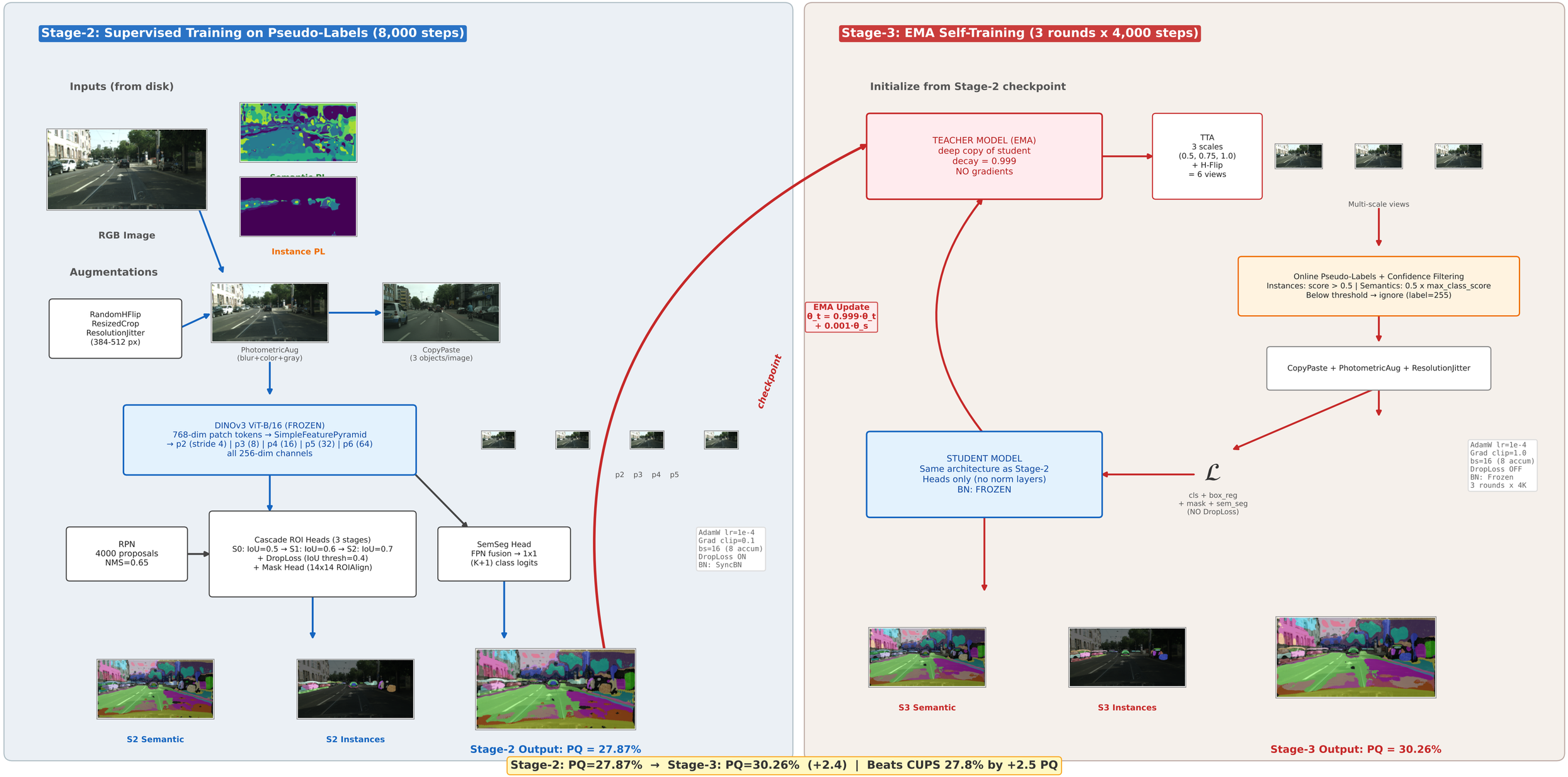

Figure 4. Network training pipeline. Stage-2 (left): Cascade Mask R-CNN with frozen DINOv3 backbone trained on pseudo-labels with DropLoss. Stage-3 (right): EMA teacher generates online pseudo-labels via TTA; student trains on confidence-filtered predictions; EMA update closes the loop. Self-training improves PQ by +2.4 points.